MIL-C-38612A(MO)
6 February 1967 M
SUPERSEDING
MIL-C-38612(USAF)
9 August 1965

MILITARY SPECIFICATION

CONTAINER, SHIPPING AND STORAGE, CNU-77/E23

1. SCOPE

1.1 This specification covers one type of metal shipping and storage container for an airborne spray tank.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-M-11473	- Magnetic-Particle Inspection, Soundness Requirements for Weldments.
MIL-W-22248	- Weldments, Aluminum and Aluminum Alloy.
MIL-L-25567	- Leak Test Compound, Oxygen Systems.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1261	- Welding Procedures for Constructional Steels.
MS18024	- Tag, Shipping.

FSC 1040

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DRAWINGS

US AIR FORCE

DL65E9008 - Container, Shipping and Storage, CNU-77/E23.

(All drawings referenced in this specification are listed on the above drawing.)

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL65E9008 or on the subsidiary drawings shall conform to the specifications listed thereon and to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the container shall conform to the specifications and drawings referenced in this document and on Drawing DL65E9008.

3.2 Assembly. The container shall be assembled as shown on Drawing 65E9008. A record shall be kept of the date of installation of the rubber shock mounts in the container.

3.3 Leakage.

3.3.1 Structual pressure. The container shall not have structual failure or permonent deformation when tested as specified in 4.3.3.1.

3.3.2 Pressure retention. The container shall not indicate a loss of more than 0.10 pound per square inch gage (psig) when tested as specified in 4.3.3.2. The result of this test shall be evaluated for compliance after 3.3.1 has been successfully completed.

3.4 Preproduction. Prior to the start of regular production, a preproduction sample of containers shall be produced in accordance with this specification for examination and test (see 4.2).

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3.4.1 Free fall drop. The container when subjected to a 12 inch gravity fall shall not crack, break, fracture or show any visible signs of failure or allow damage to the spray tank which would render it unserviceable when tested as specified in 4.2.3.1.

3.4.2 Cornerwise drop. The container when subjected to a 12 inch cornerwise gravity fall shall not crack, break, fracture or show any visible signs of failure or allow damage to the spray tank which would render it unserviceable when tested as specified in 4.2.3.2.

3.4.3 Impact. The container when subjected to an impact of 9 feet per second shall not crack, break, fracture or show any visible signs of failure or allow damage to the spray tank which would render it unserviceable when tested as specified in 4.2.3.3.

3.4.4 Roll-over. The container when subjected to forces experienced by being turned over shall not crack, break, fracture or show any visible signs of failure or allow damage to the spray tank which would render it unserviceable when tested as specified in 4.2.3.4.

3.4.5 Vacuum retention. The container when subjected to a negative differential pressure of 3.0 ± 0.5 psig shall show no indicated gain in pressure when tested as specified in 4.2.3.5. This requirement shall be evaluated for compliance after the pressure retention test (4.3.3.2) has been successfully completed.

3.4.6 Modification welding. Welding shall conform to Drawing 65E9008, subsidiary drawings and this specification. Magnetic particle inspection and acceptance standards for all steel modification welding shall be in accordance with 4.2.3.6.

3.5 Workmanship. The container shall be free from foreign matter (dirt, oil, or grease) and damage such as dents, cracks or deformation which may affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Preproduction.

4.2.1 Sample. A preproduction sample of three containers shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.2.2 Examination and nondestructive tests. The sample containers shall be examined for all requirements of the drawings and this specification and tested as specified in 4.2.3 and 4.3.3.

4.2.3 Preproduction tests. The spray tank (see 6.2(c)) shall be filled with 1,356 ± 10 pounds of water and antifreeze (to prevent freezing at normal ambient temperature at time of test) prior to placing in the container. The filled spray tank shall be properly stowed in the container before the container is tested as specified in 4.2.3.1, 4.2.3.2, 4.2.3.3 and 4.2.3.4. All valves in the container shall be blocked, prior to testing against opening at pressures within their operating range.

4.2.3.1 Free fall drop. The container shall be raised to a height of 12 ± 0.5 inch measured from the bottom of the four skids to a steel, stone or concrete surface of sufficient mass to absorb the shock without deflection, and allowed to fall freely under the force of gravity. The container shall then be leak tested at an air pressure of 4.0 ± 0.5 psig. During leakage test all welds, joints, closure flange and plumbing shall be inspected with a liquid leak detector conforming to the performance requirements of MIL-L-25567. After above tests, the container shall be inspected visually internally and externally for signs of failure in such areas (but not limited to) as fittings, permanent joints, drain plugs, hardwood skids, humidity indicator port, record receptacle, explosive container and suspension system. The container contents including spray tank shall be inspected for signs of failure; no damage shall be evident. No leakage is permitted in the spray tank and the container.

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4.2.3.2 Cornerwise drop. The container shall be supported at one corner of its base on a block 5.0 ± 0.5 inch high. A 12.0 ± 0.5 inch block shall be placed under the opposite corner of the same end of the container. The lowest point of the other end of the container shall be raised to a height of 12.0 ± 0.5 inch and allowed to fall freely onto a surface per 4.2.3.1. The test shall be applied to each of the four corners of the base. The container shall then be leak tested at an air pressure of 4.0 ± 0.5 psig. During leakage test all welds, joints, closure flange and plumbing shall be inspected with a leak detector conforming to the performance requirements of MIL-L-25567. After above tests the container shall be inspected visually internally and externally for signs of failure in such areas (but not limited to) as fittings, permanent joints, drain plugs, hardwood skids, humidity indicator port, record receptacle, explosive container and suspension system. No leakage is permitted in the spray tank and the container. The container contents including spray tank shall be inspected in accordance with 4.2.3.1.

4.2.3.3 Impact. The container shall be impacted on each end by a pendulum or equivalent method at the velocity specified in 3.4.3 against an unyielding surface. The container shall then be leak tested at an air pressure of 4.0 ± 0.5 psig. During leakage test all welds, joints, closure flange and plumbing shall be inspected with a liquid leak detector conforming to the performance requirements of MIL-L-25567. After above tests, the container shall be inspected visually internally and externally for signs of failure in such areas (but not limited to) as fittings, permanent joints, drain plugs, hardwood skids, humidity indicator port, record receptacle, explosive container and suspension system. The container contents including spray tank shall be inspected in accordance with 4.2.3.1. No leakage is permitted in the spray tank and the container.

4.2.3.4 Roll-over. The container shall be tipped slowly from its erect and normal shipping attitude until it falls freely and by its own weight on its side. The container shall be tipped in this manner for a 360° cycle in the direction from side to top to side to normal position. The container shall then be leak tested at an air pressure of 4.0 ± 0.5 psig. During leakage test all welds, joints, closure flange and plumbing shall be inspected with a liquid test detector conforming to the performance requirements of MIL-L-25567. After above tests, the container shall be inspected visually internally and externally for signs of failure in such areas (but not limited to) as fittings, permanent joints, drain plugs, hardwood skids, humidity indicator port, record receptacle, explosive container and suspension system. The container contents including spray tank shall be inspected in accordance with 4.2.3.1. No leakage is permitted in the spray tank and the container.

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4.2.3.5 Vacuum retention. The container shall be evacuated to a negative differential pressure of 3.0 ± 0.5 psig below atmospheric pressure. There shall be no more than 0.10 psig indicated gain in pressure over the 24 hour period after stabilization is attained. A man/machine error correction of ± 0.05 psig shall be applied to the computed gain before determining whether the pressure gain meets the requirements.

4.2.3.6 Welding. Welding acceptance shall be in accordance with MIL-STD-1261 and as follows:

(a) No cracks allowed -- (figures 1, 2, and 3 of MIL-M-11473).

(b) Fusion -- Combined total lack of fusion shall not exceed 1 inch in each 6 inches of weld (figure 4 of MIL-M-11473).

(c) Undercut -- Combined total of undercut shall not exceed 1-1/2 inch in each 6 inches of weld (figure 5 of MIL-M-11473).

(d) Penetration -- Combined total of incomplete penetration shall not exceed 1 inch in each 6 inches of weld.

(e) Porosity -- Scattered porosity shall not exceed the pattern of plate 11 of MIL-W-22248.

4.2.4 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.2.2 to be acceptable.

4.3 Regular production.

4.3.1 Lotting. A lot shall consist of the containers except M11 canisters produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions.

4.3.2 Sampling for examination and nondestructive tests. Each container shall be examined and tested as specified in 4.3.3 One container shall be taken at random from each month's production and examined and tested as specified in 4.2.3 and 4.3.3.

4.3.3 Tests. All valves shall be blocked during tests against opening at pressures within their operating range.

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4.3.3.1 Structual pressure. The container shall be subjected to an air pressure of 2.0 + 0.5 psig and all seams, joints, plumbing and welds checked for leakage with a liquid leak detector conforming to the performance requirements of MIL-L-25567. The pressure shall be increased to 5.0 + 0.5 psig and held for a maximum period of 1 minute and then decreased to 3.0 + 0.5 psig. The container shall be checked for leakage with a liquid leak detector. The container shall then be subjected slowly to a pressure of 8.0 ± 0.5 psig for a minimum period of 1 minute. The container may leak at the closure seal only. The pressure shall be decreased to 5.0 ± 0.5 psig and the container checked for leakage with a liquid leak detector. There shall be no leakage except as noted above.

4.3.3.2 Pressure retention. The container shall be subjected to an air pressure of 4.0 ± 0.5 psig. The container pressure shall be measured for 24 hours after stabilization of pressure and temperature by means of gages, manometers or other suitable means which are capable of recording values to the nearest 0.05 psig (0.10 inch of mercury). The ambient temperature and container pressure shall be recorded. The container air pressure and temperature shall again be recorded and the pressure corrected for any temperature or barometric pressure change after the above test is completed. There shall be no more than 0.10 psig loss of pressure indicated over the 24 hour period after stabilization is attained and a man/machine error correction of ± 0.05 psig to the computed pressure loss is applied.

4.3.4 Acceptance/rejection criteria. The container shall comply with the examination and tests specified in 4.3.2 to be acceptable (see 6.4). The sample container taken at random from each month's preproduction shall comply with the examinations and tests specified in 4.2.2 to be acceptable (see 6.4). When a random sample container selected from the month's production fails any of the tests specified in 4.3.2, the remaining containers of the lot shall be sampled and examined in accordance with MIL-STD-105, level S-2, using an AQL of 2.5 percent defective for the defects that contributed to test failure.

5. PREPARATION FOR DELIVERY

5.1 Packing. Not applicable.

5.2 Marking. In addition to any special marking required by the contract or order, shipping and storage containers shall be marked in accordance with Drawing 65E9008. A shipping tag conforming to MS18024 shall be attached by wiring to one of the lifting brackets. The tag shall note the consignee and consignor's name and address (see 6.2(d)).

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6. NOTES

6.1 Intended use. The container covered by this specification is intended for use as a domestic and overseas shipping and storage container for the filled TMU-28/B spray tank and accessories.

6.2 Ordering data. Procurement documents should specify the following:

(a) *Title, number and date of this specification.*
(b) Preproduction.

(1) Time allowed for supplier submission of any samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping insturctions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

(c) Number of spray tanks (Dwg 63J56706) and accessories to be supplies by the Government and date to be delivered.
(d) Shipping destination.

6.3 Modification. The container covered by this specification is a modified J57 jet engine shipping and storage container. Work on the container modification, as specified on drawings cited in this specification, is described in T.O. 35E20-3-1-2, Modification Procedure.

6.4 Rejected containers. Rejected containers may be reworked to Government approved procedures and resubmitted for testing.

Custodian:
Army - MU

Review interests:
Air Force - 18

Preparing activity:
Army - MU (EA)

Project No. 1040-A067

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119 R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will ensure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM APR 65 1426 REPLACES NAVSHIPS FORM 4853, WHICH IS OBSOLETE

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